



Introduction to MEMS

An insight into the world miniaturized mechanical and electro-mechanical devices. <http://digit.in/nov20-35>



Flavanol to lower blood pressure

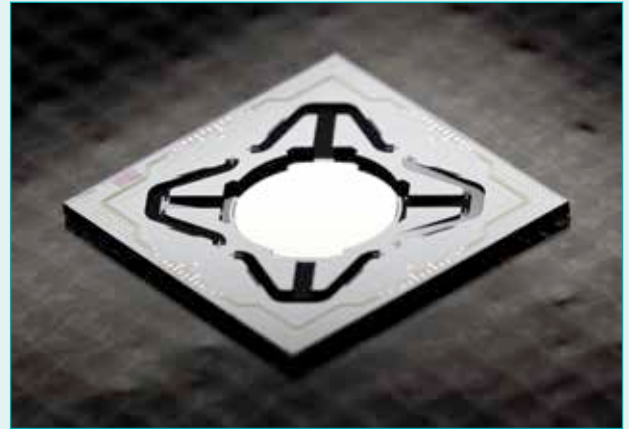
A plant based nutrient called flavanol, found in apples, tea and nuts has shown to lower BP levels. <http://digit.in/nov20-36>



MEMS

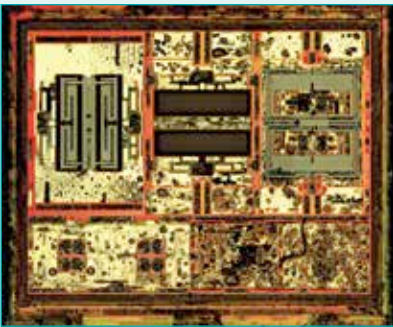
Microelectromechanical devices, or MEMS are tiny devices typically ranging from 0.02 to 1.0 mm in size, with microchips, sensors and even mechanical components. The components are even smaller.

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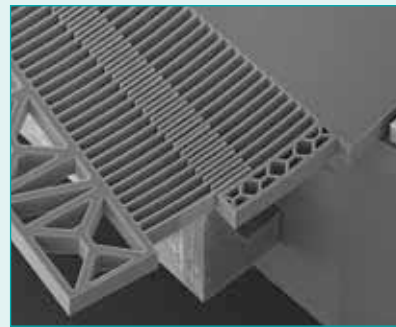
Credit: Fraunhofer IPMS

↑ This is a microscanner actuator used for modulating light beams, measuring 500 micrometers across. There are two types of microscanners, rotational and translational, and the one shown here is a translational microscanner, which deflects the incidental light beam.



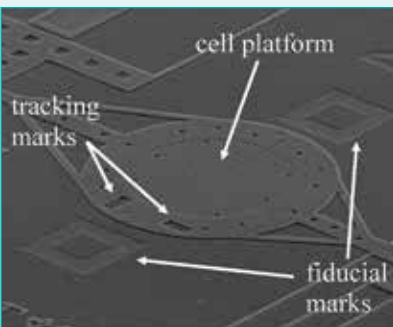
Credit: Teptobars

← A die that forms part of a microscale gyroscope and accelerometer, measuring 2782x2718 micrometers. Conventional digital logic is located below this die.



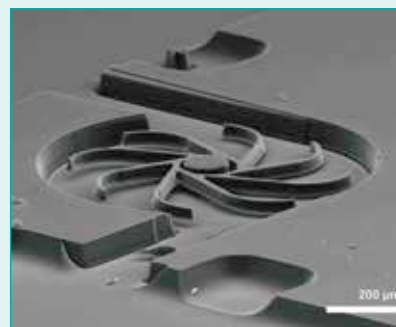
Credit: FBK, MT Lab

← Demonstration micromechanical structures produced by etching, these kinds of chips can be used for working with microfluids and to make a wide range of really tiny sensors.



Credit: National Institute of Standards and Technology

← Shown here is a device known as a "cell puller", where a living cell sticks to the circular platform in the middle, 200 micrometers diameter, after which its mechanical properties are observed.



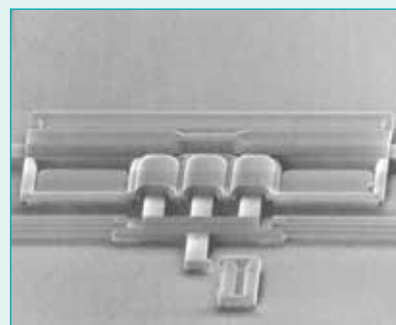
Credit: Hungarian academy of sciences

← This is a demonstration of silicon micromachining and proton beam writing, which is a form of lithography that allows granular control over the fabrication of shapes and forms.



Credit: Denis Elbl, KIT

← This is a micro-mechanical optical switch, where really tiny mirrors are used to reflect beams of light that can then be used for various purposes downstream.



Credit: Sandia National Laboratory

← This is a tiny steam engine with three pistons. An application of current heats water that pushes the pistons outwards. Capillary action pulls the pistons back into place.